



Fig. 4: The potentiometer integration into the circuit

mode by shorting pins 1-2, as indicated in the PCB layout, the RC timing network is connected to the DIAC. In this configuration, the circuit operates as a conventional AC phase-angle controller using the high-power, snubberless BTA41-1200BW TRIAC.

In this mode, the 250kΩ-500kΩ linear potentiometer, together with R1 (10kΩ) and C1 (0.1μF/400V polyester, non-polarised), forms an adjustable RC timing network. During each half-cycle of the AC mains, capacitor C1 charges through the potentiometer. When the voltage across C1 reaches the breakover voltage of DIAC D1, typically around 32V for a DB3 or equivalent device, the DIAC conducts abruptly and generates a sharp trigger pulse at the TRIAC gate.

As a result, the BTA41-1200BW TRIAC is triggered at a variable phase angle during each half-cycle, depending on the potentiometer setting. Delayed triggering allows only a small portion of the sine wave to reach the load, producing a lower RMS output voltage and reduced power. Earlier triggering increases the conduction angle, allowing a larger portion of the sine wave to reach the load and raising the RMS voltage to nearly the full mains value. This provides smooth and continuous control of the AC output voltage.

A snubber network, consisting of R2 (100Ω) in series with C3 (100nF) across the TRIAC MT1-MT2 terminals, suppresses voltage transients and prevents false triggering. This improves circuit reliability, particularly when driving inductive loads.

In dimmer mode, the MOC3041 opto-triac driver is bypassed, enabling true random-phase control. This configuration is suitable for smooth dimming of resistive loads such as lamps and heaters, as well as for speed control of universal motors, including fans. With appropriate heat sinking, the BTA41-1200BW can safely handle loads of up to 40A RMS, corresponding to approximately 9kW at 230V AC.

## Working in solid-state relay mode

When the solder jumper configuration is changed to relay mode by connecting JP1 pins 2-3 and enabling JP2, the TRIAC gate drive is routed through the MOC3041 optocoupler. In this mode, the optocoupler can be driven directly by a microcontroller or an external logic signal, providing complete electrical isolation between the low-voltage control circuitry and the high-voltage AC load.

In solid-state relay mode, the TRIAC functions as an on/off electronic switch. This configuration is suitable for microcontroller-based automation, industrial control systems, and high-power AC switching applications.

For heavy loads, the heat sink specified in the parts list must be used. Wiring should be selected according to the expected load current to ensure safe and reliable operation.

## Testing

After completing the PCB assembly and verifying all connections, configure the solder jumper for voltage controller (dimmer) mode by shorting pins 1 and 2 of JP1. Ensure that pins 2 and 3 remain open if they were previously connected. Connect a lamp or another suitable AC load to the output terminals, either via the connector marked 'Load' or the pins labelled C/L, as applicable.

Rotate the potentiometer to vary the output voltage and the power

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delivered to the load. For calibration, a voltmeter may be connected across the load to observe the RMS output voltage while adjusting the potentiometer. Reference positions may be marked on the rotary knob for commonly used voltage levels, enabling quick, repeatable control of lamp brightness, heater power, or motor speed during subsequent operations.

For solid-state relay (on/off) mode, desolder pins 1 and 2 of JP1, then short pins 2 and 3, and short JP2 if provided to enable the optocoupler path. In this configuration, apply a low-voltage DC control signal, typically 3V to 12V, to the optocoupler input via connector H1 (signal and GND). The TRIAC then operates as a solid-state switch, turning the connected load fully on or fully off without any dimming action.

The 1200V rating refers to the maximum blocking, or off-state, voltage that the TRIAC can safely withstand without breakdown. This rating provides a safety margin against mains surges and transient spikes, where peak voltages can momentarily exceed 800-1000V in high-voltage environments.

This rating does not indicate that the circuit outputs or regulates 1200V RMS using the potentiometer. The circuit is intended to control standard AC mains voltages, typically 0-230-260V AC, as used in domestic and industrial power systems. **EFY**

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